

How the measurement works

From sentences to a number, step by step

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August 2026

The setup: one grid of sentences

We take **300 sentences** and their professional translations into **41 languages**.
Sentence 7 in German means the same thing as sentence 7 in Tamil.

The model reads each one and produces a vector. That gives a grid:

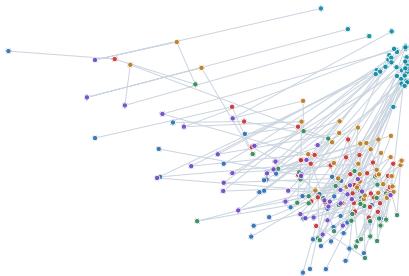
	sentence 1	sentence 2	...	sentence 300
English	$h_{1,1}$	$h_{1,2}$		$h_{1,300}$
German	$h_{2,1}$	$h_{2,2}$		$h_{2,300}$
⋮				
Tamil	$h_{41,1}$	$h_{41,2}$		$h_{41,300}$

12,300 vectors. **Every question we ask is a question about this grid.**

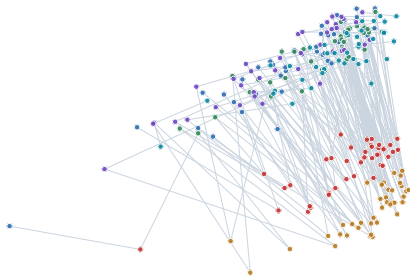
What the grid actually looks like

Each dot is one sentence; grey lines join translations of the same sentence.
The clusters are languages, not meanings.

Qwen3-4B language share 0.24



bloom-1b7 language share 0.61



If the model represented *meaning* and ignored *language*, every grey line would shrink to a point: all 6 translations of a sentence would land in the same place. Instead the dots group by colour. **The model's geometry is organised by language first.**

Two things move a vector

Where a vector sits is driven by exactly two systematic factors, plus noise:

$$\underbrace{h_{\ell,c}}_{\text{one cell}} = \underbrace{\mu}_{\text{grand average}} + \underbrace{a_c}_{\substack{\text{which sentence} \\ \text{(meaning)}}} + \underbrace{b_\ell}_{\substack{\text{which language} \\ \text{(language)}}} + \underbrace{e_{\ell,c}}_{\text{the rest}}$$

- a_c is shared by *all 41 translations* of sentence c — it is the column effect
- b_ℓ is shared by *all 300 sentences* in language ℓ — it is the row effect

This is a standard two way analysis of variance, applied per dimension of the representation and then summed.

Separating them: a worked example

Take one dimension, 2 languages, 3 sentences:

	s1	s2	s3	row mean
English	1.0	3.0	5.0	3.0
German	2.0	4.0	6.0	4.0
column mean	1.5	3.5	5.5	3.5

- **Meaning effects** = column means – grand mean = $-2, 0, +2$ (spread out)
- **Language effects** = row means – grand mean = $-0.5, +0.5$ (tight)
- Residual = cell – row – column + grand = 0 everywhere

Meaning moves these vectors much more than language does. That is a **low language share** — the good case.

The number

$$\text{language share} = \frac{\sigma_{\text{language}}^2}{\sigma_{\text{language}}^2 + \sigma_{\text{meaning}}^2}$$

0	language does nothing; all 41 translations of a sentence land together
1	meaning does nothing; all 300 sentences in a language land together

- Measured across 33 models the value runs from **0.59 to 0.90**
- Language identity is the *dominant* organising factor in every model tested, including the best ones
- This is the quantity LFS reported, and it is a real and useful measurement

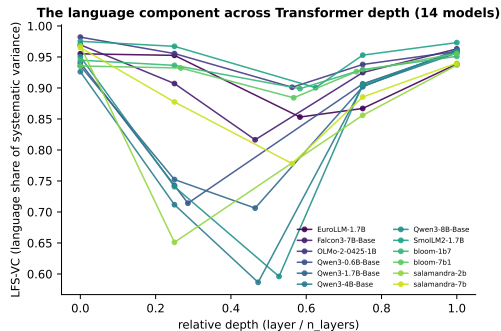
It changes with depth, in the same way everywhere

Measured at every layer of 31 models, the shape is always the same:

- **high** at the input embedding — tokens are language specific
- **lowest** around the middle to late layers — the most language neutral point
- **high again** at the output — the model must commit to words in one language

What differs between models is **how deep the dip goes**: 0.38 for Mistral-Nemo, 0.48 for Llama-3.1, but only 0.87 for Falcon3-3B.

All other readings are taken at each model's own minimum.



Where reporting it as one number goes wrong

Take the same worked example and **shrink everything by a factor of 10** — the model keeps its structure but flattens it:

	s1	s2	s3	row mean
English	0.10	0.30	0.50	0.30
German	0.20	0.40	0.60	0.40
column mean	0.15	0.35	0.55	0.35

Meaning effects: $-0.2, 0, +0.2$. Language effects: $-0.05, +0.05$.

$$\frac{c\sigma_{\text{language}}^2}{c\sigma_{\text{language}}^2 + c\sigma_{\text{meaning}}^2} = \text{exactly the same number}$$

Both parts shrank, so the share cannot move. The model lost almost all its structure and the score is unchanged.

This is not hypothetical

A trained variant lost **95 percent** of its meaning structure at the measurement layer — and posted the **best score in the study**.

- Confirmed in closed form on controlled distortions: the share moves less than a millionth while meaning structure falls 99 percent
- Any measure written as a share of a total inherits this
- The fix is not a better estimate of the share. It is to **report the parts, not the ratio**

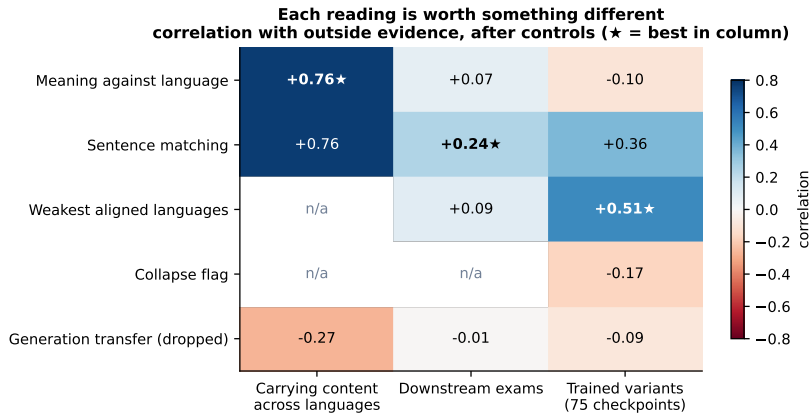
What RMFS reports instead

Four readings, per language, side by side:

1. **Meaning against language** — the decomposition above, unchanged. Answers: is the space organised by content or by language?
2. **Sentence matching** — for each sentence in language ℓ , is its English translation the nearest of the 300 candidates? Answers: can the model find equivalent content across languages?
3. **Weakest aligned languages** — the same matching restricted to the worst aligned tail. Answers: how bad is it where it is worst?
4. **Collapse flag** — meaning structure compared against a reference checkpoint. Answers: did training destroy structure?

The first is a property of the whole grid; the second and third are properties of individual rows against the English row; the fourth needs a before and after.

Why four and not one



Each reading was checked against three different kinds of outside evidence. **No reading wins two columns.** They measure genuinely different things, so averaging them into one number throws away the information that makes them useful.

Turning readings into a score

Each reading is on its own scale, so they are converted before being combined:

- | | |
|-------------------------------------|---|
| 1. compute the four readings | per language, at the model's own layer |
| 2. convert each to a percentile | against a fixed reference grid of 33 models |
| 3. average the four percentiles | giving 0 to 100 per language |
| 4. take the median across languages | giving one number per model |

- A score of **74** means better than 74 percent of the reference models
- Percentiles rather than raw values, because averaging raw scales is precisely what let a collapse hide
- About four GPU minutes per model

Reading a result

	meaning vs language	sentence matching	weakest languages	collapse flag
a healthy model	moderate	high	moderate	1.0
the damaged variant	<i>moderate</i>	<i>high</i>	moderate	0.05
a poorly aligned model	low	low	low	1.0

- The damaged variant is invisible on the first three readings and obvious on the fourth
- A poorly aligned model is the opposite pattern
- **The pattern across the four readings is the diagnosis**; the single score is only a summary for ranking